Computational Statistical Physics
Exercise 5

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1 Description

This report will attempt to tackle the problem of simulating N interacting particles inside a 3D cubic geometry. The focus will mainly lie in implementing accurate boundary conditions and numerical schemes such that the system adheres to physical principles, in terms of total energy conservation. Specifically for this problem, a two-body interaction potential will be considered

$$V_{ij} = V(\vec{r}_i, \vec{r}_j) = V\left(|\vec{r}_i - \vec{r}_j|\right) = V(r_{ij})$$

Which formula follows as

$$V(r) = 4\left[\left(\frac{1}{r}\right)^{12} - \left(\frac{1}{r}\right)^{6}\right]$$

and may be recognized as the Lennard-Jones potential, extensively used in the context of molecular dynamics simulations. In this case, the constants have been set to 1 as mainly the qualitative behavior will be of later interest.

Given this smooth potential, (see fig. (1)) an event driven simulation is not applicable here. Instead, each particle must be updated at every time-step using an appropriate numerical scheme. In this case, the position is updated using the Verlet integrator

$$\vec{x}_i(t_{n+1}) = 2\vec{x}_i(t_n) - \vec{x}_i(t_{n-1}) + \Delta t^2 \vec{a}_i(t_n) + \mathcal{O}(\Delta t^4)$$

Also noting here that

$$\vec{a}_i = -\frac{1}{m}\nabla V(r_{ij}) = -\frac{1}{m}\frac{\vec{r}_i - \vec{r}_j}{r_{ij}}\frac{\partial V(r_{ij})}{\partial r}$$

The method being fourth order, numerical accuracy is possible even in the steep region of the potential. Finally, acknowledging that each particle update has to take into account the position of the $N-1$ other particles when calculating the potential gradient, each and every update step is on the order of $\mathcal{O}(N^2)$.

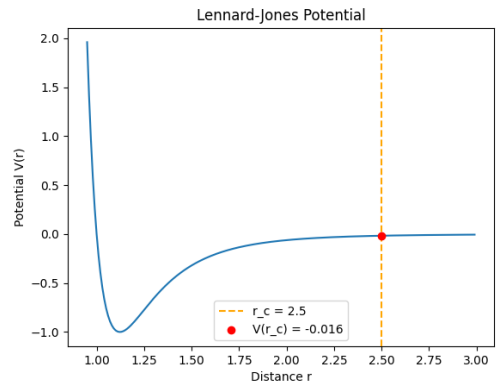

Figure 1: Lennard Jones potential and cut-off radius $r_c = 2.5$.

The true order can be lowered by introducing a cut-off radius r_c, beyond which no potential or gradients are calculated.

Introducing a cut-off radius $r_c = 2.5$, introduces a discontinuity $\Delta V|_{r=r_c} = 0.016$. This discontinuity is dealt with by adjusting the entire potential with a magnitude $V(r_c)$, such that the following formula is assumed

$$V'(r) = \begin{cases} V(r) - V(r_c) & : r < r_c \\ 0 & : r > r_c \end{cases}$$

2 Simulation Setup

For this simulation, a cubic geometry of $L \times L \times L$ is considered. With $r_c/2$ being roughly the radius of each particle, it is off interest to yield an approximation in terms of how many particles can at most fit inside the geometry. A reasonable approach would be to first consider each particle as a cube of side-length r_c. Then L/r_c such cubes can be stacked along each dimension, yielding a result

$$N_{cubic} = \left(\frac{L}{r_c}\right)^3$$

Considering the particular particle geometries in more detail, namely spherical shapes, it is also trivially realized that the particles can be stacked in a hexagonal pattern. See the illustration below for a 2D case

Aligning layers of 2D hexagonal surfaces composed of spheres such that the third dimension also adheres to the hexagonal pattern, while considering a box large enough such that the gaps at the edges resulting from the hexagonal lining can be neglected, it is possible to fit at most the following number of particles

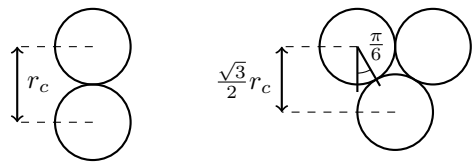

Figure 2: Left: Circles stacked in a cubic pattern. Right: Circles stacked hexagonicaly, contracting each axis by a factor $\sqrt{3}/2$.

$$N_{hexagonal} = \frac{8}{3\sqrt{3}} \left(\frac{L}{r_c} \right)^3 \approx 1.54 N_{cubic}$$

Letting $r_c = 2.5$, it follows $N_{hexagonal} = 8/(3\sqrt{3}) \cdot 4^3 = 98.53$. As such, 98 can be considered as an absolute upper threshold for the number of particles, given box and particle sizes. However, to let particles move freely a particle number $N = 30$ is chosen. In terms of position, the particles are initialized carefully such that no two particles are overlapping in a nonphysical way, causing energy spikes during the first couple of time-steps. The particle velocities are also initialized in a way resembling a physical system, in this case a Maxwell-Boltzmann distribution where the generated total momentum is normed to zero. Finally, for each simulation performed, all particle masses are set to 1. The only parameters varying between simulations are the timesteps Δt and the average initial particle kinetic energy $\langle \epsilon_0 \rangle$. The relationship between $\langle \epsilon_0 \rangle$ and the velocity initialization is illustrated below

$$\vec{v}_{0,i} = v_{0,i} \cdot \hat{n}_i : \begin{cases} \hat{n}_i \sim \mathcal{U}(\mathbb{S}^2) \\ \frac{1}{2} m v_{0,i}^2 = \epsilon_{0,i} \sim \text{Exp}(\frac{1}{\langle \epsilon_0 \rangle}) \end{cases}$$

3 Total Energy Conservation

For each timestep, the kinetic and potential energy of each particle is summed to yield the total energy of the system. Given that the simulation aims to resemble a physical system, it is of great interest to validate that the simulated system adheres perhaps one of the most important physical principles - energy conservation under no external influence. The system is simulated for a total time $T = 10$ with step-sizes in the range $\Delta t \in \{0.010, 0.005, 0.001\}$. Storing the total system energy at each timestep, it is possible to compare the energy conservation with respect to discretization errors. Such is done in the following plot

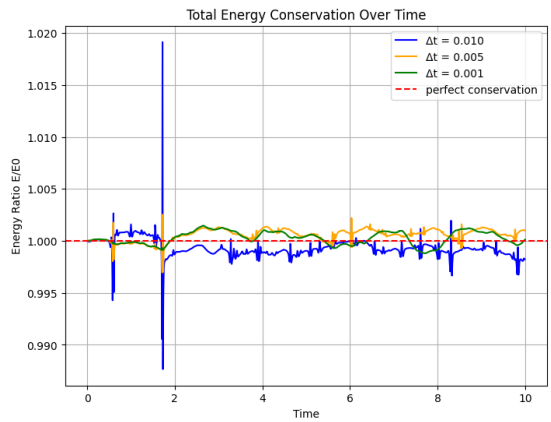

Figure 3: Energy conservation with respect to discretization. Initial particle energies $\langle \epsilon_0 \rangle = 10$.

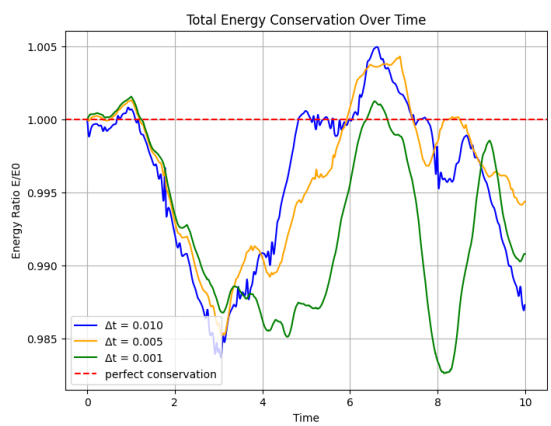

Figure 4: Energy conservation with respect to discretization. Initial particle energies $\langle \epsilon_0 \rangle = 1$.

Plotting the energy conservation for varying timesteps, it is clearly noticeable that the the step-size plays an important role in reducing large energy spikes. These energy spikes result from particle collisions and not boundary crossovers. They are thereby a result of the potential energy interactions between the particles and not their kinetic energies. This can be verified by comparing fig. (3) against fig. (4), where the initial particle kinetic energies have been lowered by a factor 10. In the latter simulations, the particles do not approach each other as closely. As such, large potential spikes are not prevalent, although the system instead suffers from total energy drift. The drift seems to be largely independent of step-size.

4 Particle Energy Fluctuations

The following part will deal with investigating particle energy fluctuations and how they relate to the discretization and initial kinetic energies. With ϵ_i now denoting the total energy for each particle, the energy fluctuations are now defined as follows

$$\Delta\epsilon = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\epsilon_i - \langle\epsilon\rangle)^2}$$

These fluctuations are calculated at every timestep and plotted in a similar manner to the total energy with respect to time.

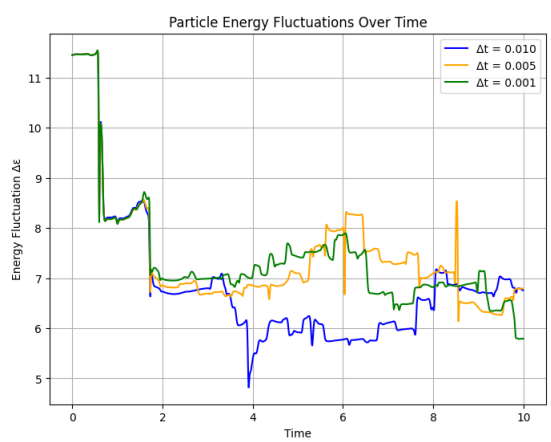

Figure 5: Particle energy fluctuations with respect to discretization. Initial particle energies $\langle\epsilon_0\rangle = 10$.

In both cases, the fluctuations start fairly high, before they rapidly decrease and stabilize around $t = 2$. In the plot with higher initial kinetic energies, this

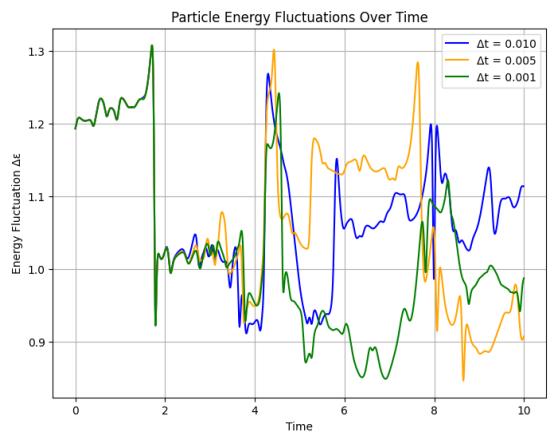

Figure 6: Particle energy fluctuations with respect to discretization. Initial particle energies $\langle \epsilon_0 \rangle = 1$.

stabilization is more pronounced. The time-interval $t \in [0, 2]$ might be considered as the equilibration time during which the particles approach their "true" energy distribution, from having initially followed the Maxwell-Boltzmann distribution discussed previously. Contrary to what one might expect given the energy jumps in the conservation plots for larger step-sizes, the step-size seems to have a negligible effect on the particle energy fluctuations.